

Project Title:

AI API Marketplace & Gateway (AI-Exchange)

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GitHub Repository:

<https://github.com/Sharan-Muthu-Krishna/AI-Exchange-API-Marketplace>

Live Demo:

- **Frontend:** <https://aiexchange-frontend.onrender.com>
- **Backend API:** <https://aiexchange-backend.onrender.com/docs>

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1. Introduction

The rapid growth of artificial intelligence has led to the creation of powerful AI models for tasks such as text analysis, image recognition, fraud detection, and recommendation systems. However, the AI ecosystem lacks a centralized and secure platform where:

- **Vendors** can publish and monetize AI APIs
- **Buyers** can discover, subscribe to, and consume AI services
- **Platforms** can securely manage authentication, billing, and usage

AIExchange addresses this gap by providing a **production-grade SaaS platform** that functions as both an **AI API marketplace** and a **secure API gateway**.

Unlike simple API directories, AIExchange:

- Proxies all requests through a controlled gateway
- Tracks usage at the API-call level
- Implements credit-based billing and commission logic
- Prevents direct exposure of vendor endpoints

2. Project Objective

The primary objectives of AIExchange are:

- To design and build a **multi-tenant SaaS marketplace** for AI APIs
- To implement a **secure API gateway pattern** that abstracts vendor services
- To provide **credit-based billing with real-time usage tracking**
- To support **role-based access control** for Admin, Vendor, and Buyer users

3. System Overview

AIExchange consists of three core components:

3.1 Frontend (Next.js)

- React-based UI using TypeScript
- Role-specific dashboards
- Public marketplace for API discovery
- Clean, modern SaaS-style design

3.2 Backend (FastAPI)

- Async REST API using Python
- JWT authentication and API key validation
- API gateway with proxy routing
- Billing, usage tracking, and analytics

3.3 Database (PostgreSQL)

- Cloud-hosted PostgreSQL (Neon)
- Normalized relational schema
- Async access using asyncpg

4. Architecture & Design

4.1 API Gateway Pattern

AIExchange acts as a **reverse proxy gateway**:

1. Buyer sends a request to `/v1/apis/{api_id}/run`
2. Gateway validates:
 - API key authenticity
 - Subscription status
 - Credit balance
3. Request is forwarded to the vendor's private endpoint
4. Response is returned to the buyer
5. Usage and billing are recorded

Benefits of Gateway Architecture

- Vendor endpoints remain private
- Centralized security and authentication
- Consistent usage tracking and billing
- Platform-level commission enforcement

4.2 Multi-Tenancy

- Vendors are isolated from one another
- Buyers have independent wallets and API keys
- All usage and billing is tracked per tenant

5. Technology Stack

Layer	Technology
Frontend	Next.js (React + TypeScript)
Backend	FastAPI
Database	PostgreSQL (Neon Cloud)
ORM	SQLAlchemy
Authentication	JWT + bcrypt
API Keys	SHA-256 hashed
Rate Limiting	SlowAPI
Deployment	Render
Version Control	Git & GitHub

6. Database Schema

Core Tables

Table	Description
users	All users with roles
vendors	Vendor profiles and wallets
buyers	Buyer profiles and wallets
apis	Vendor-published APIs
api_keys	Secure API keys
subscriptions	Buyer-API mapping
api_usage	Per-call usage logs
wallet_transactions	Credit movements
platform_revenue	Platform commission

7. Authentication & Security

7.1 JWT Authentication

- Passwords hashed with bcrypt
- Access token expiry: 30 minutes
- Refresh token expiry: 7 days
- Algorithm: HS256

7.2 API Key Security

- Secure random key generation
- Stored only as hashed values
- Shown once during creation
- Validated on every request

7.3 RBAC

Role	Permissions
Admin	Full platform access
Vendor	API management, earnings
Buyer	API usage, subscriptions

8. Billing & Credit System

- Credit-based usage model
- 10% platform commission
- Vendor receives remaining credits

9. How to run the system

1. Backend

```
cd backend
python -m venv venv
venv\Scripts\activate

pip install -r requirements.txt
python init_db.py
uvicorn app.main:app --reload
```

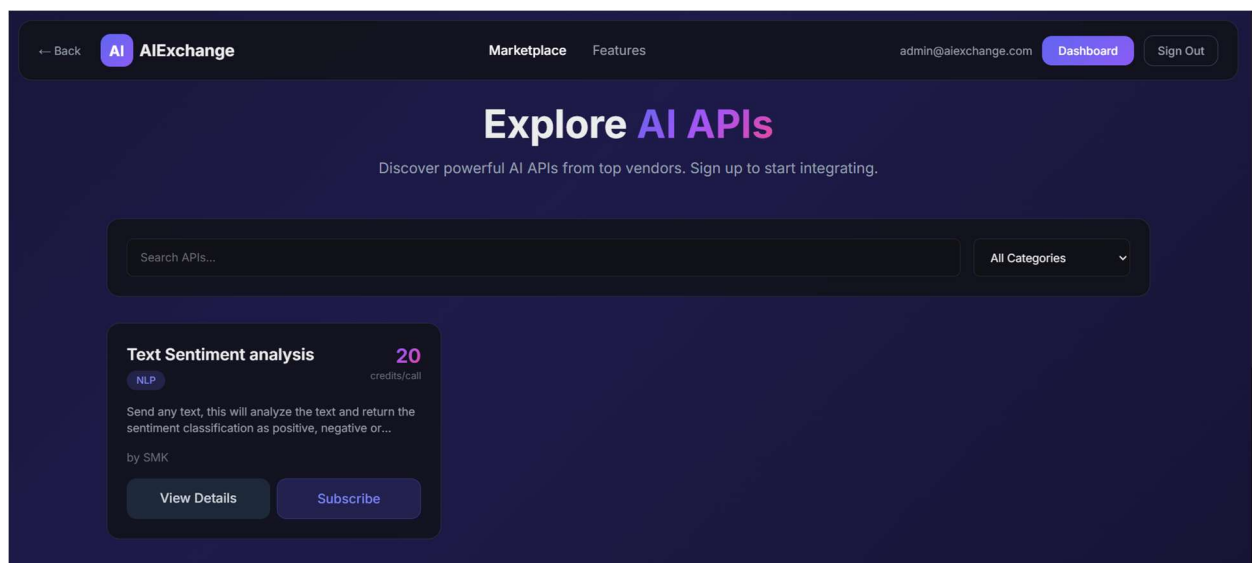
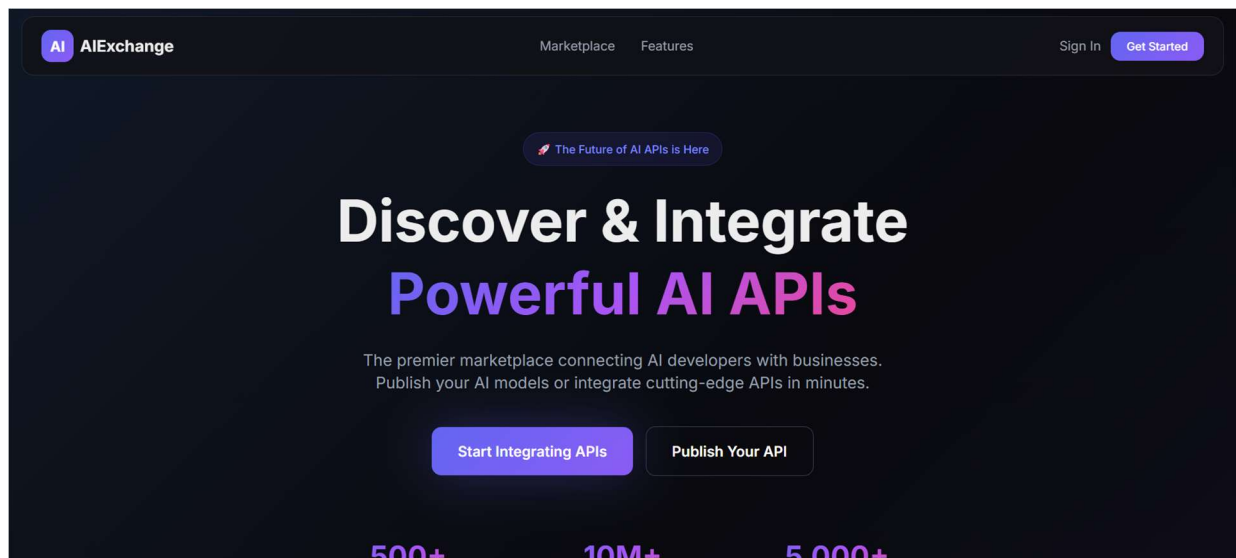
2. Frontend

```
cd frontend  
npm install  
npm run dev
```

10. Screenshots

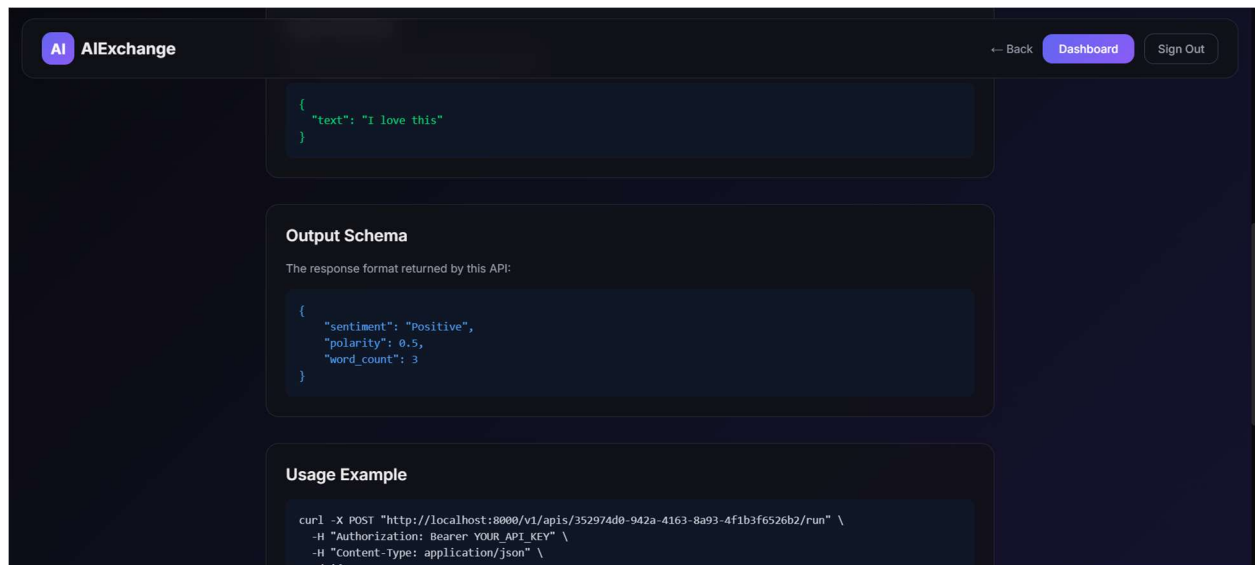
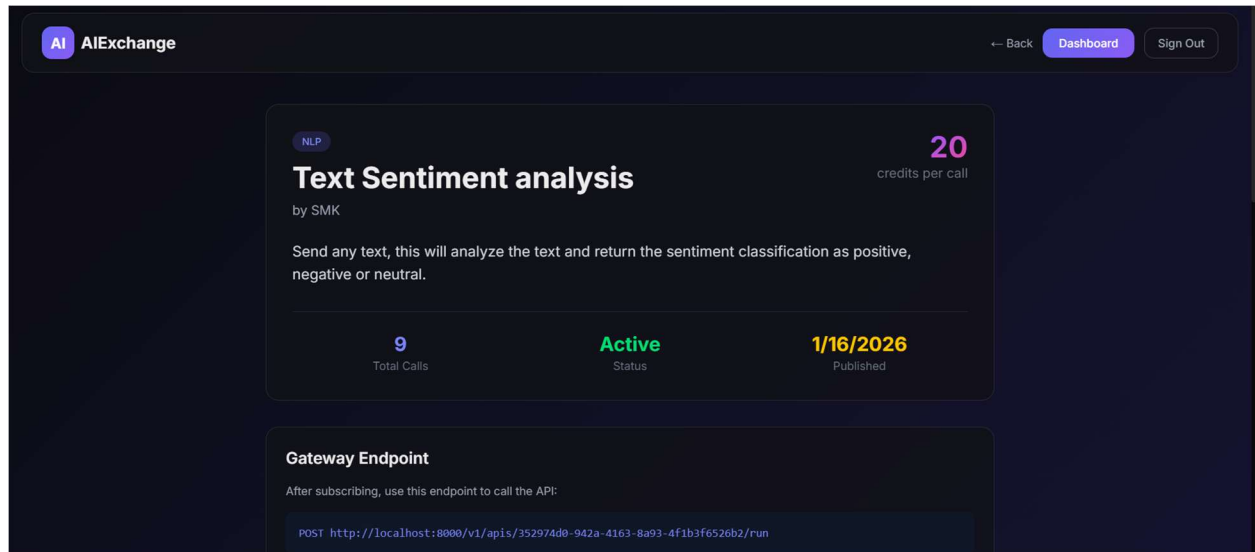
10.1 Landing Page & API Marketplace

The landing page introduces **AIExchange** as an AI API Marketplace and Gateway. It highlights the platform's purpose and allows users to browse the **API Marketplace**, where all available AI APIs are listed with their categories, pricing, and vendors. Users can explore APIs publicly before signing up.



10.2 API Details & Documentation Page

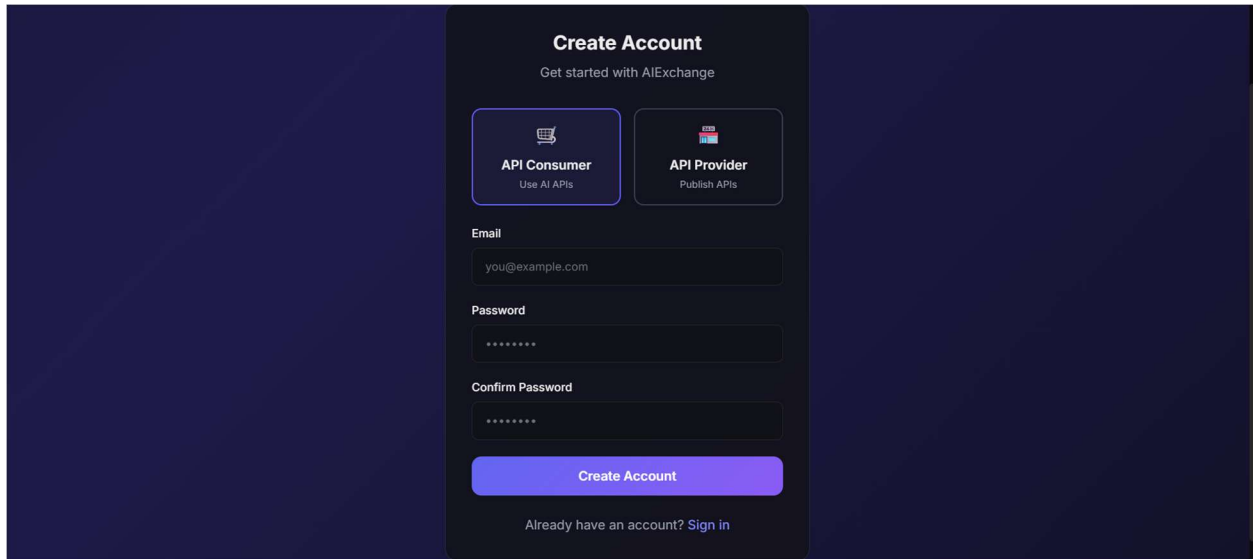
This page displays complete documentation for a selected API, including: API description and use case, Input request schema, Output response schema, Pricing per API call. It helps buyers clearly understand how to integrate and use the API before subscribing.



10.3 Signup Page

The signup page allows users to register either as a **Vendor** or a **Buyer**.

Based on the selected role, the platform provisions role-specific access and redirects users to the appropriate dashboard after authentication.



The 'Create Account' form is centered on a dark blue background. It features two role selection buttons: 'API Consumer' (Use AI APIs) and 'API Provider' (Publish APIs). Below these are input fields for Email, Password, and Confirm Password. A prominent 'Create Account' button is at the bottom, with a link to 'Sign in' for existing users.

Create Account
Get started with AIExchange

API Consumer
Use AI APIs

API Provider
Publish APIs

Email
you@example.com

Password

Confirm Password

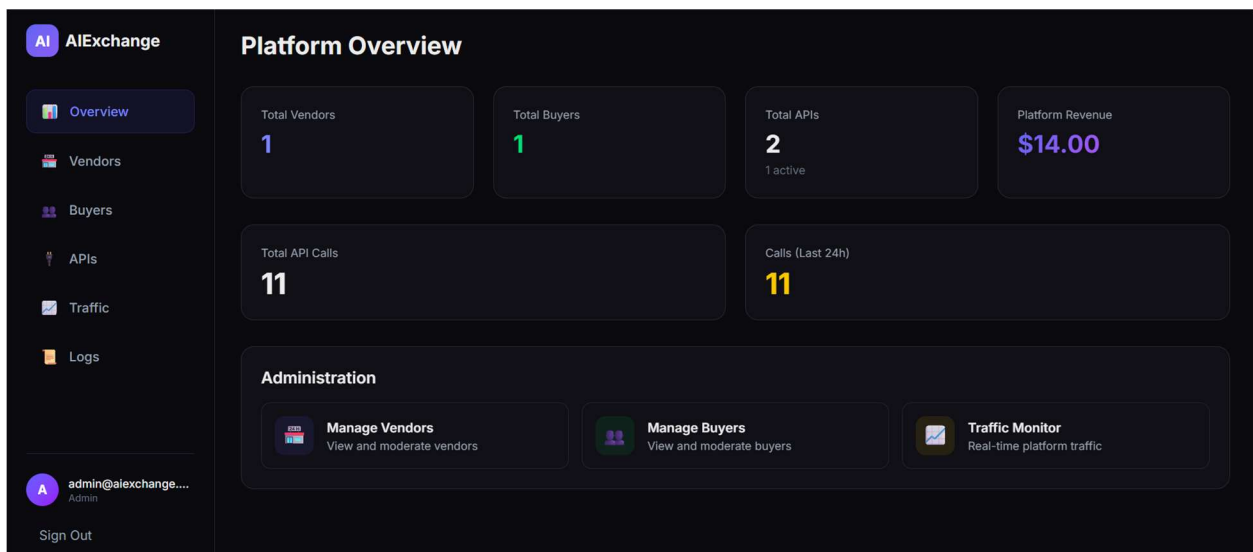
Create Account

Already have an account? [Sign in](#)

10.4 Admin Dashboard (APIs & Traffic Logs)

The Admin dashboard provides a **system-wide view** of the platform.

Admins can monitor all registered users, published APIs, overall traffic, API usage logs, and platform activity, enabling oversight, analytics, and abuse detection.



AI

AIExchange

Overview

Vendors

Buyers

APIs

Traffic

Logs

admin@aiexchange....

Admin

Sign Out

https://aiexchange-frontend-render.com/admin/vendors

APIs

AllActiveInactive

NAME	CATEGORY	PRICE	STATUS	CREATED	ACTIONS
Text Sentiment analysis	NLP	20 credits	Active	1/16/2026	Disable
Text Sentiment analysis	NLP	20 credits	Inactive	1/16/2026	Enable

AI

AIExchange

Overview

Vendors

Buyers

APIs

Traffic

Logs

admin@aiexchange....

Admin

Sign Out

Traffic Monitor

24h48h72h168h

Total Calls

15

Success Rate

73.3%

Failed Calls

4

Avg Response Time

1600.07ms

Top APIs

1

Text Sentiment analysis

9 calls

2

Text Sentiment analysis

6 calls

AI

AIExchange

Overview

Vendors

Buyers

APIs

Traffic

Logs

admin@aiexchange....

Admin

Sign Out

API Usage Logs

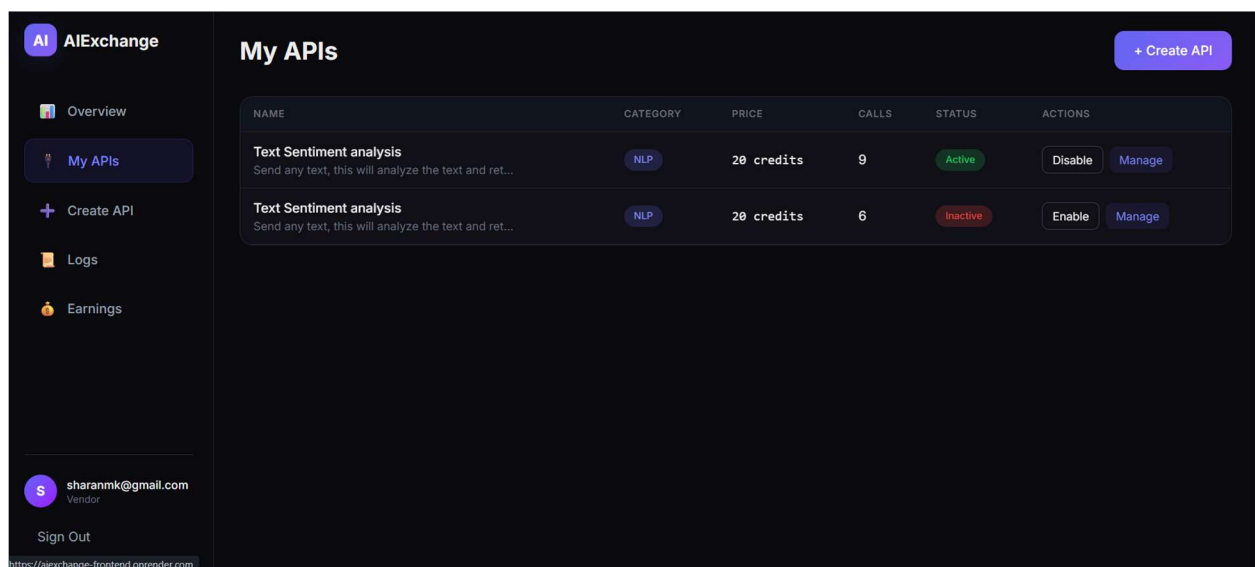
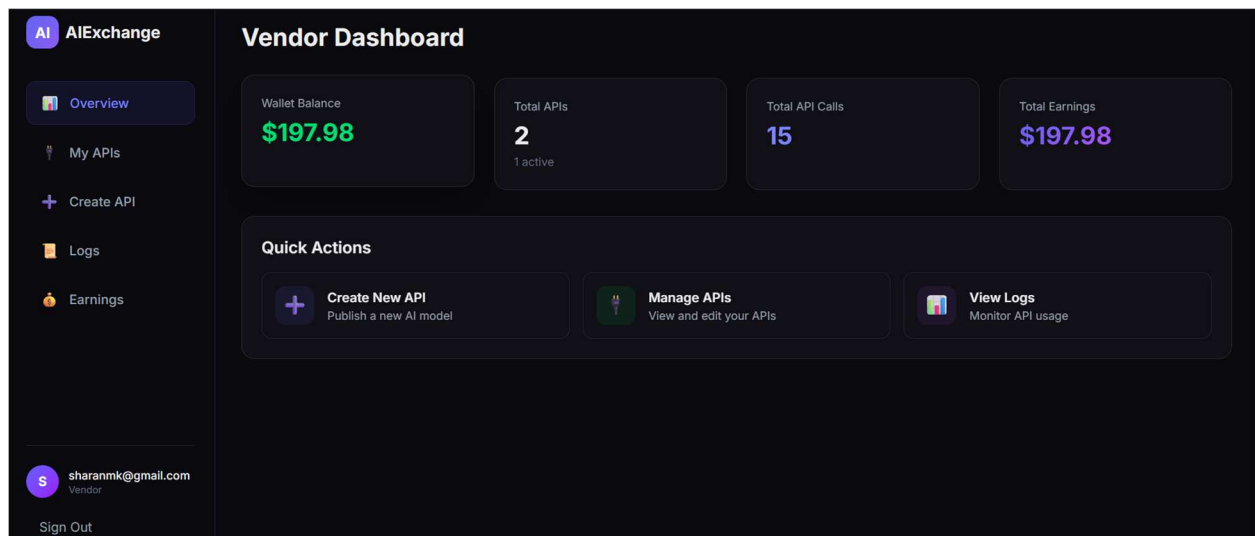
TIMESTAMP	API	BUYER	STATUS	CREDITS	RESPONSE TIME
1/16/2026, 11:59:59 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	SUCCESS	20.00	17120ms
1/16/2026, 11:29:03 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	SUCCESS	20.00	839ms
1/16/2026, 8:19:52 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	SUCCESS	20.00	197ms
1/16/2026, 8:18:32 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	FAILED	0.00	310ms
1/16/2026, 8:17:32 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	FAILED	0.00	224ms
1/16/2026, 8:16:58 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	FAILED	0.00	329ms
1/16/2026, 8:16:53 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	FAILED	0.00	311ms
1/16/2026, 7:54:27 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	SUCCESS	20.00	1019ms
1/16/2026, 7:45:37 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	SUCCESS	20.00	962ms
1/16/2026, 7:39:22 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	SUCCESS	19.99	202ms
1/16/2026, 7:33:34 AM	Text Sentiment analysis	sharanmkbuyer@gmail.com	SUCCESS	19.99	1659ms

10.5 Vendor Dashboard (My APIs & API Logs)

The Vendor dashboard allows API providers to:

- Create and manage their published APIs
- View API usage statistics
- Analyze request logs and success rates
- Track earnings generated from API usage

This dashboard enables vendors to manage their services independently.



API Usage Logs					
TIMESTAMP	API	STATUS	CREDITS	RESPONSE TIME	ERROR
1/16/2026, 12:01:44 PM	Text Sentiment analysis	SUCCESS	20.00	251ms	-
1/16/2026, 12:01:42 PM	Text Sentiment analysis	SUCCESS	20.00	157ms	-
1/16/2026, 12:01:34 PM	Text Sentiment analysis	SUCCESS	20.00	255ms	-
1/16/2026, 12:01:33 PM	Text Sentiment analysis	SUCCESS	20.00	165ms	-
1/16/2026, 11:59:59 AM	Text Sentiment analysis	SUCCESS	20.00	17120ms	-
1/16/2026, 11:29:03 AM	Text Sentiment analysis	SUCCESS	20.00	839ms	-
1/16/2026, 8:19:52 AM	Text Sentiment analysis	SUCCESS	20.00	197ms	-
1/16/2026, 8:18:32 AM	Text Sentiment analysis	FAILED	0.00	310ms	Vendor API returned status 502
1/16/2026, 8:17:32 AM	Text Sentiment analysis	FAILED	0.00	224ms	Vendor API returned status 429
1/16/2026, 8:16:58 AM	Text Sentiment analysis	FAILED	0.00	329ms	Vendor API returned status 429
1/16/2026, 8:16:53 AM	Text Sentiment analysis	FAILED	0.00	311ms	Vendor API returned status 429

10.6 Buyer Dashboard (Subscriptions & Credits)

The Buyer dashboard shows:

- Subscribed APIs
- API keys generated for programmatic access
- Credit wallet balance and usage history

Buyers can manage subscriptions, track spending, and purchase additional credits from this interface.

AIExchange

Overview

Marketplace

Subscriptions

API Keys

Credits

sharanmkbuyer@gm...
Buyer

Sign Out

Dashboard

Credit Balance

80.02

+ Add Credits

Active Subscriptions

2

Total API Calls

15

API Keys

2

Quick Actions

Browse Marketplace

Discover new APIs

Manage API Keys

Create or revoke keys

Purchase Credits

Top up your balance

AI

AIExchange

Overview

Marketplace

Subscriptions

API Keys

Credits

S

sharanmkbuyer@gm...

Buyer

Sign Out

My Subscriptions

API	GATEWAY URL	PRICE	STATUS	ACTIONS
Text Sentiment analysis	/v1/apis/352974d8-942a-4163-8a93-4f1b3f6526b2/run	20 credits	Active	Cancel
Text Sentiment analysis	/v1/apis/c2fa2617-ed5b-43a8-a6d9-898aac6a4878/run	20 credits	Cancelled	

AI

AIExchange

Overview

Marketplace

Subscriptions

API Keys

Credits

S

sharanmkbuyer@gm...

Buyer

Sign Out

Credits

Current Balance

180.02

credits

Total Spent

219.98

Purchase Credits

50 credits

100 credits

250 credits

500 credits

1000 credits

100

Purchase 100 Credits

This is a simulated purchase for demonstration purposes.

Recent Transactions

DATE	DESCRIPTION	TYPE	AMOUNT
1/16/2026, 12:06:58 PM	Credit purchase: 100.0 credits	PURCHASE	+100.00
1/16/2026, 12:01:44 PM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 12:01:43 PM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 12:01:34 PM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 12:01:33 PM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 11:59:59 AM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 11:29:44 AM	Credit purchase: 100.0 credits	PURCHASE	+100.00
1/16/2026, 11:29:03 AM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 11:28:16 AM	Credit purchase: 100.0 credits	PURCHASE	+100.00
1/16/2026, 8:19:52 AM	API call: Text Sentiment analysis	DEBIT	-20.00

AI

AIExchange

Overview

Marketplace

Subscriptions

API Keys

Credits

S

sharanmkbuyer@gm...

Buyer

Sign Out

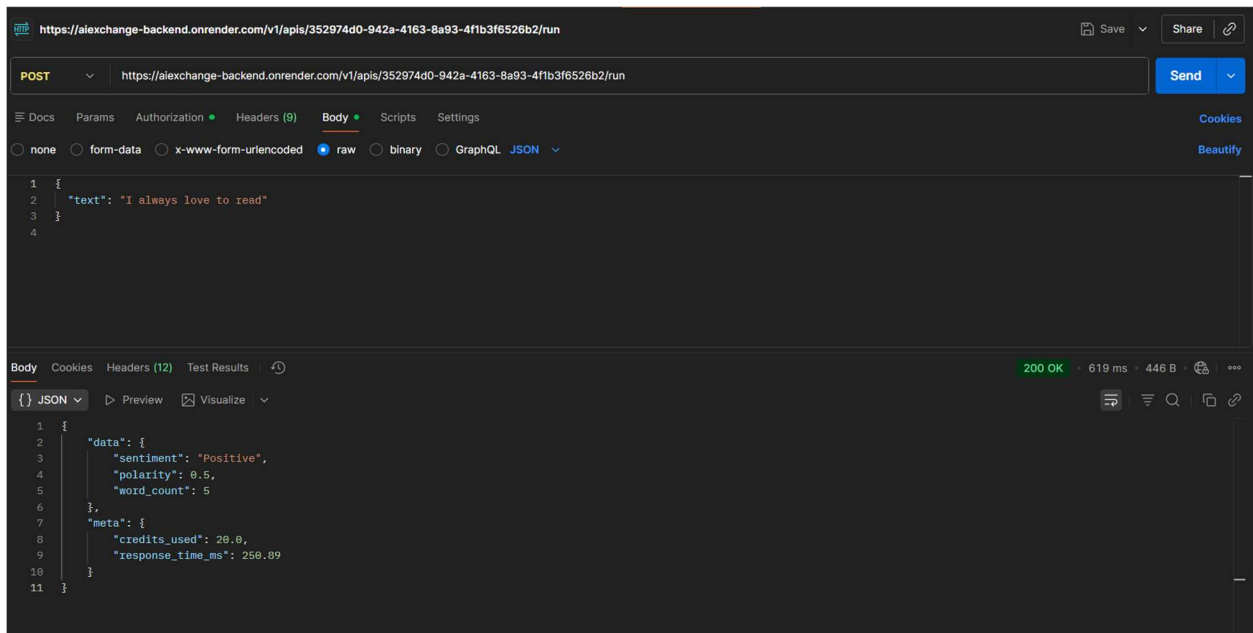
Recent Transactions

DATE	DESCRIPTION	TYPE	AMOUNT
1/16/2026, 12:06:58 PM	Credit purchase: 100.0 credits	PURCHASE	+100.00
1/16/2026, 12:01:44 PM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 12:01:43 PM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 12:01:34 PM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 12:01:33 PM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 11:59:59 AM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 11:29:44 AM	Credit purchase: 100.0 credits	PURCHASE	+100.00
1/16/2026, 11:29:03 AM	API call: Text Sentiment analysis	DEBIT	-20.00
1/16/2026, 11:28:16 AM	Credit purchase: 100.0 credits	PURCHASE	+100.00
1/16/2026, 8:19:52 AM	API call: Text Sentiment analysis	DEBIT	-20.00

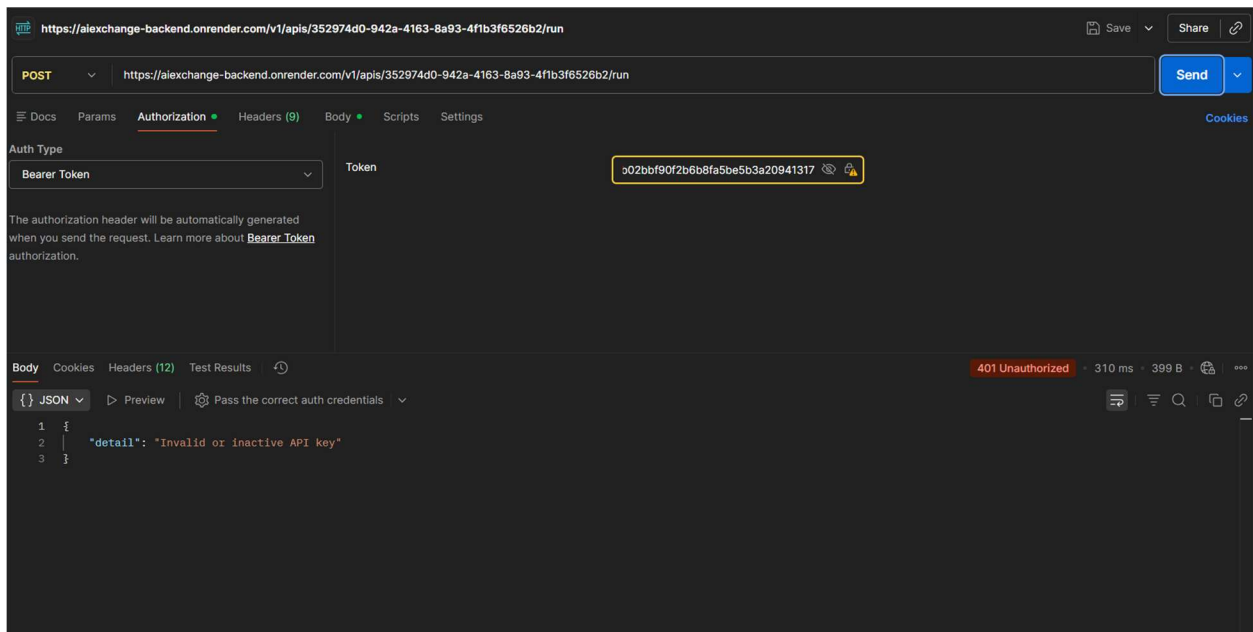
10.7 API Testing via Postman

This screenshot demonstrates testing an API using **Postman**.

A valid API key generated by the platform is passed in the Authorization header. The request is routed through the AIXchange Gateway, authenticated, billed, and forwarded to the vendor's endpoint before returning the response.



Below is the output for invalid API key:



11. Conclusion

AIExchange demonstrates the design and implementation of a **full-scale SaaS API platform** with:

- Secure authentication and authorization
- API gateway architecture
- Usage-based billing simulation
- Multi-tenant marketplace design
- Cloud deployment and CI/CD

This project strongly reflects **backend engineering, platform design, SaaS architecture, and cloud deployment skills**.